

On the Use of the ubcdiss Template

by

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On the Use of the ubcdiss Template

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Abstract

This document provides brief instructions for using the `ubcdiss` class to write a UBC-conformant dissertation in $\text{\LaTeX}$. This document is itself written using the `ubcdiss` class and is intended to serve as an example of writing a dissertation in $\text{\LaTeX}$. This document has embedded Unique Resource Locators (URLS) and is intended to be viewed using a computer-based Portable Document Format (PDF) reader.

Note: Abstracts should generally try to avoid using acronyms.

Note: at University of British Columbia (UBC), both the Graduate and Postdoctoral Studies (GPS) Ph.D. defence programme and the Library's online submission system restricts abstracts to 350 words.

This document was typeset in `gpscopy` mode.

Lay Summary

The lay or public summary explains the key goals and contributions of the research/scholarly work in terms that can be understood by the general public. It must not exceed 150 words in length.

Preface

At UBC, a preface may be required. Be sure to check the GPS guidelines as they may have specific content to be included.

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Glossary

This glossary uses the handy `acronym` package to automatically maintain the glossary. It uses the package's `printonlyused` option to include only those acronyms explicitly referenced in the $\text{\LaTeX}$ source. To change how the acronyms are rendered, change the `\acsfont` definition in `diss.tex`.

GPS Graduate and Postdoctoral Studies

PDF Portable Document Format

URL Unique Resource Locator, used to describe a means for obtaining some resource on the world wide web

Acknowledgments

Thank those people who helped you.

Don't forget your parents or loved ones.

You may wish to acknowledge your funding sources.

Methods

Citizen science program

The Tree Spotters was a citizen science program that started in 2015 and aimed to train citizen scientists for accurate and rigorous phenological monitoring at the Arnold Arboretum of Harvard University (42.30 °N, -71.12 °W) (Figure ??). From 2015 to 2024, hundreds of citizen scientists monitored 77 individual trees and shrubs of 14 species. They regularly followed those individuals from budburst in the spring to leaf colouring in the fall using the National Phenology Network (NPN) phenophases [2]: Leaves (483), Colored leaves (498), Fruits (516), Ripe Fruits (390), Falling leaves(471), Recent fruit or seed drop (504), Increasing leaf size (467), Breaking leaf buds (371), Flowers or flower buds (500), Open flowers (501), Pollen release (502). Not all phenophases were recorded for every tree, for every year, and some trees miss several years of data.

Phenological monitoring and sample collection

From 20 to 22 April 2025, we collected tree ring samples only for the tree species that were previously monitored for phenology (excluding *Fagus grandifolia* because of beach bark disease concerns) for a total of 49 individuals (Table S1). For this, we collected two 5-mm diameter cores, 15-cm length at 1.3 meters above ground, using an increment borer (Mora Borer, Haglöfs Sweden, Bromma, Stockholm, Sweden). We collected the cores perpendicular to the slope and at 180 degrees from each other, cleaning the increment borer with alcohol (70% ethanol) and the inside with a brush before collecting each core. We stored the cores in modified plastic straws to allow aeration and were left to dry at ambient temperature for three months.

Sample processing, imaging and measuring

We mounted the cores on wooden mounts, sanded them using progressively finer sandpaper grits (150, 300, 400, 600, 800, 1000), and then scanned the cores at a

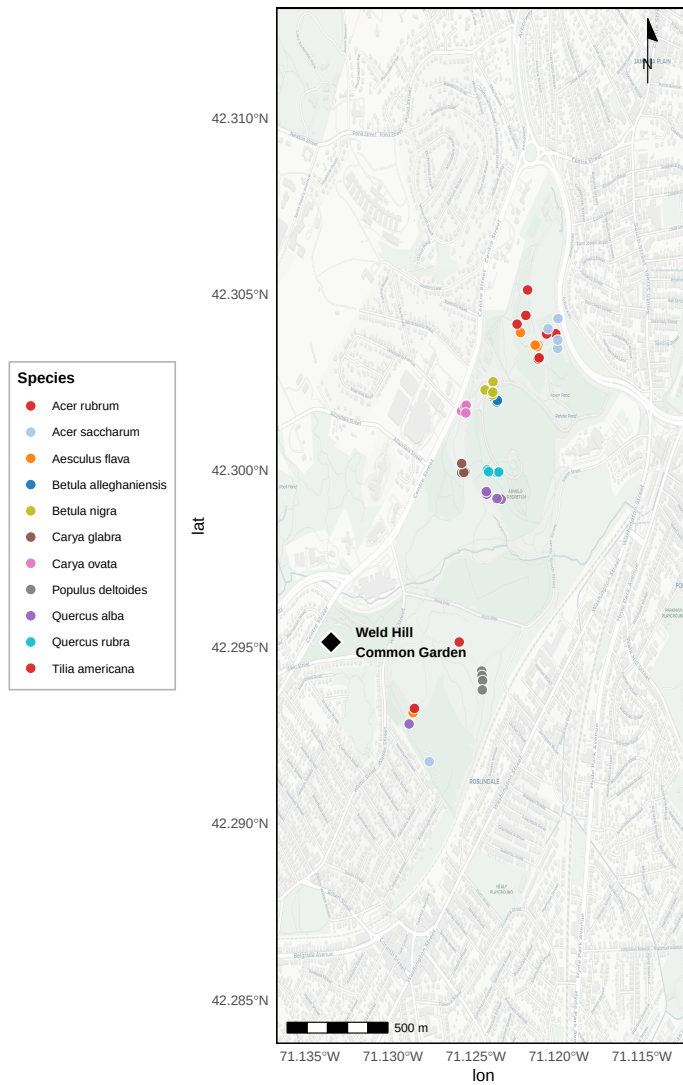


Figure 1: Locations of the trees in the arboretum. The colors of the dots match the model estimates of Figure ??.

resolution of 6250 dpi with a high-resolution treering scanner (Fong, unpublished). Using the digitized images, we measured ring widths with ImageJ [4]

We performed visual cross dating. We did not perform statistical crossdating because of the short chronologies that limit the capacity of these analyses [3], and

for similar reasons, we did not detrend the ring widths. Visual analyses of trends showed no consistent allometric patterns (Figure S1); thus we included only a year intercept to correct for potential effects of different years, not otherwise included in our model.

Data analyses

To understand how growth shifted with season lengths we used linear Bayesian hierarchical models, estimating ring width with differing season lengths using four different season predictors (GDD, GSL, SOS and EOS). We calculated the thermal season using growing degree days (GDD) (mean daily temperature above 5°C) accumulated between leafout and budset, and the calendar season (or growing season length (GSL)) as the number of days between leafout and budset. Finally, we used leafout and budset as the start (SOS) and end of the season (EOS) as separate predictors that represent the boundaries of the growing season.

We used the daily climate data from the climate station located at Weldhill research station for 2018 and 2019 (<https://arboretum.harvard.edu/research/data-resources/weather/>). Because the data was not available for that station after August 2020, we used the daily climate data from the weather station “Boston (Weld Hill)” of the Network for Environment and Weather Applications (NEWA) for the whole year in 2020 (<https://newa.cornell.edu/all-weather-data-query/>).

For each predictor, we estimated its effect on ring width observed for each tree in each year as:

$$\begin{aligned} \ln(\text{ringwidth})_i &\sim N(\mu, \sigma_y) \\ \mu &= \alpha + \alpha_{\text{species}} + \alpha_{\text{prov}} + \alpha_{\text{treeid}} + \alpha_{\text{year}} + \beta_{\text{species}} \times \text{season predictor} \\ \alpha_{\text{prov}} &\sim N(0, \sigma_{\alpha_{\text{prov}}}) \\ \alpha_{\text{treeid}} &\sim N(0, \sigma_{\alpha_{\text{treeid}}}) \end{aligned}$$

where i denotes each observation, α represent intercepts and β represent slopes.

This model yields a species-specific effect of each seasonal predictor (e.g. GDD) while also allowing other variation due to species ($\alpha_{species}$), alongside variation due to provenance (α_{prov}), tree id (α_{treeid}), year (α_{year}) and a species specific growth rate ($\beta_{species}$). We chose weakly informative priors for all models; increasing the priors by 2x did not change the results trends. We ran four chains for a total of 8000 warmup (discarded) and 8000 sampling iterations, which we kept for posterior distribution estimates. The models did not have any divergent transitions, had a $\hat{R}$ below 1.01, and values of effective sample size (n_{eff}) greater than 10% of the model iterations. We wrote our models using Stan [1] with the rstan package version 2.32.7 [5] to run the Stan code in R.

As a supplementary analyses, we ran the models using basal area increment (BAI), as the response variable. For this, we summed each ring width separately at the three distinct locations on the cross sections yielding three radii from pit to bark. We then calculated the area of each ring by subtracting the radius from the inner ring from the outer ring: $\pi \times (\text{Inner radius}^2 - \text{Outer radius}^2)$. Then, we averaged across the three BAI for each ring and fit the models with the four different predictors.

We also fit the models for SOS and EOS using the full vs restricted datasets.

Results

Discussion

Supplemental

Supplemental Methods

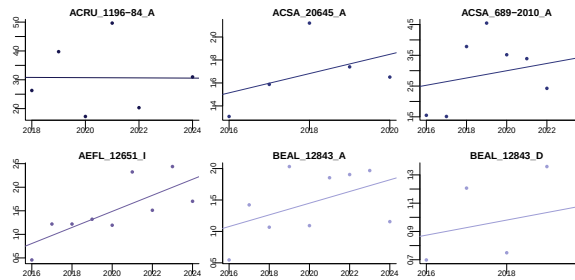


Figure S1: We randomly sampled 20 trees from our dataset and represent the ring width as a function of year to represent the allometry patterns across all years we had ring width for.

Table S1: Treepotters species grouped by tree type, life history, and wood anatomy. A total of 49 individual trees were sampled.

Common Name (Latin)	Life History Strategy	Wood Anatomy	<i>n</i>
American basswood (<i>Tilia americana</i>)	Fast-growth, moderate life	Diffuse-porous	5
Eastern cottonwood (<i>Populus deltoides</i>)	Fast-growth, short life	Diffuse-porous	4
Northern red oak (<i>Quercus rubra</i>)	Moderate-growth, long life	Ring-porous	4
White oak (<i>Quercus alba</i>)	Slow-growth, long life	Ring-porous	5
Pignut hickory (<i>Carya glabra</i>)	Slow-growth, long life	Ring-porous	4
Shagbark hickory (<i>Carya ovata</i>)	Slow-growth, long life	Ring-porous	5
River birch (<i>Betula nigra</i>)	Fast-growth, short life	Diffuse-porous	4
Yellow birch (<i>Betula alleghaniensis</i>)	Moderate-growth, moderate life	Diffuse-porous	4
Sugar maple (<i>Acer saccharum</i>)	Slow-growth, long life	Diffuse-porous	5
Red maple (<i>Acer rubrum</i>)	Slow-growth, long life	Diffuse-porous	4
Yellow buckeye (<i>Aesculus flava</i>)	Moderate-growth, moderate life	Diffuse-porous	5

Supplemental Results

Bibliography

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- [5] Stan Development Team. RStan: the R interface to Stan, 2025. URL <https://mc-stan.org/>. → page 4

Appendix A

Supporting Materials

This would be any supporting material not central to the dissertation. For example:

- additional details of methodology and/or data;
- diagrams of specialized equipment developed.;
- copies of questionnaires and survey instruments.